Phase Changes and Heating Curves

Energy, temperature and changes of state

Name _____ Class _____ Date _____

DATA FOR WATER

c(ice) = 2.09 J/(g·°C) · c(water) = 4.184 J/(g·°C) · c(steam) = 2.01 J/(g·°C) · M = 18.02 g/mol

ΔH(fus) = 6.01 kJ/mol at 0 °C · ΔH(vap) = 40.7 kJ/mol at 100 °C

Ethanol: M = 46.07 g/mol · ΔH(vap) = 38.6 kJ/mol

Use q = mcΔT for temperature changes and q = nΔH for phase changes. Convert everything to kJ before adding.

A. Ideas

1. Name the phase change, and say whether it is endothermic or exothermic: (a) dew forming on grass (b) ice cubes shrinking in a freezer (c) a puddle drying (d) frost forming on a window (e) wax solidifying

2. Explain why the temperature of a pot of boiling water stays at 100 °C while the burner is still on.

B. One-step calculations

3. Calculate (a) the heat needed to melt 100. g of ice at 0 °C (b) the heat released when 50.0 g of water freezes at 0 °C.

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4. Calculate the heat needed to vaporize 10.0 g of ethanol at its boiling point.

C. Multi-step calculations

5. Calculate the heat needed to turn 10.0 g of ice at −20.0 °C into liquid water at 50.0 °C.

6. A drink is cooled by adding 200. g of ice at 0 °C. Calculate the heat the ice absorbs as it melts and the meltwater warms to 20.0 °C.

D. Vapour pressure and boiling

7. At the top of Mount Everest the air pressure is about one third of an atmosphere, and water boils at about 70 °C. Explain why, and why cooking pasta there takes longer.

E. Challenge

8. Compare the heat released to skin at 37 °C by 10.0 g of steam at 100 °C and by 10.0 g of water at 100 °C. Use your answer to explain why steam burns are so severe.

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